Name:			

Embedded Software - Monthly Examination - Question cum Answer Booklet

Instructions:

A) All Questions are to be answered. For each question, marks are indicated inside square brackets [ ].

B) All answers to be written in the space provided below the Question; use reverse side of the paper for rough work.

C) If needed use additional sheets of paper (white paper) for rough work and tie all of them to the question paper.

E) You are not allowed to take any additional sheets out of the examination hall.

F) **Calculators should NOT be used** **Max Marks: 40, Max Time: 120 Mins**

==========

C – Programming - Section **[M:40, T:120]**

Interview Questions:

1. Explain the usage of header files in C programming?

2. Answer the following
   (a) What is the difference between #include <stdio.h> and #include "stdio.h"? When would you use each?
   (b) Write a macro MAX(a, b) that returns the larger of two values. Ensure it works correctly when used as: `int x = MAX(10, 20) ;

3. What is the output of the below code.
```
#include <stdio.h>
int main()
{
char flag=0x0f;
 flag &= ~0x02;
printf("%d",flag);
 return 0;
}
```
   Answer:

4. What is the operator used to make 1's One's compliment?

5. What is the output of this C code?
```
 #include
   int main()
   {
      if (~0 == 1)
```

```
        printf("yes\n");
      else
        printf("no\n");
    }
```

6. Write a program to display number of ‘1’’ and ‘0’ in an integer data. If given data is 0x1289 then display how many numbers of ‘1’ and ‘0’.

7. Write a program to flip all the bits in each integer. If given data is 0x05 then output should be 0xfa. If given data ix 0x0005 then output should be 0xfffa .

8. Which compilation unit is responsible for adding header files content in the source code?

9. What is the output

```
#define x 10 * 8
#include <stdio.h>
int main()
{
   int a = x * x+1;
   printf("%d",a);
   return 0;
}
```

10. Write a C program to print even numbers from 1 to 250 using a do-while loop.
11. What is the output of the following code? Explain the role of static in the variable count

```
#include <stdio.h>

void increment(void)

 {
   static int count = 0;
   count++;
   printf("%d ", count);
}
int main(void)
{
   increment();
   increment();
   increment();
   return 0;
}
```

12. Predict the output.

```
#include <stdio.h>
int func(int n) {
   if (n <= 0)
      return 0;
   printf("%d ", n);
   return n + func(n - 1);
}

int main(void) {
   printf("\nSum = %d", func(3));
   return 0;
}
```

13. Given two files:

file1.c:

```
#include <stdio.h>
int x = 10;

void print_x(void) {
   printf("In print_x: %d\n", x);
}
```

file2.c:

```
#include <stdio.h>
extern int x;

int main(void) {
   printf("In main: %d\n", x);
   x = 20;
   print_x();  // Assume print_x is linked from file1.c
   return 0;
}
```

14. What is the output of the following code?

```
#include <stdio.h>
int main(void) {
   int i = 0;
   while (i < 5) {
      printf("%d ", i++);
   }
   printf("\n");
```

```
    i = 0;
    while (i < 5) {
        printf("%d ", ++i);
    }
    return 0;
}
```

15. Write a short C program that demonstrates the difference between: -

- An automatic local variable – A
- Static local variable

Add brief comments explaining their scope and behaviour.

16.
```
#include <stdio.h>

int main(void)

 {

    int i = 1;

    printf("%d %d %d\n", i, ++i, i++);

    return 0;

}
```

- Is the output of this program well-defined according to the C language standard?
- Explain your answer in detail with respect to how many times ”i” is modified between sequence points, and why different embedded compilers may print different results.

17. Consider an embedded target where unsigned char is 8 bits.

```
#include <stdio.h>
int main(void)
{
    unsigned char i;
    for (i = 250; i <= 255; i++)
     {
       printf("%u ", i);
    }
    return 0;
}
```

Identify the issue with this code, does this loop ever terminate on such target?

18. Consider sizeof(int) == 4 and sizeof(int *) == 4.

```
void inspect(int arr[10])
{
   printf("In inspect: %zu %zu\n",
      sizeof(arr),
     sizeof(arr) / sizeof(arr[0]));
}
int main(void)
{
   int a[10] = {0};
   printf("In main  : %zu %zu\n",
         sizeof(a),
         sizeof(a) / sizeof(a[0]));

   inspect(a);
   return 0;
}
```

- What exactly is printed by this program? (Write both lines.)
- Explain why sizeof(arr) inside inspect does not give the same result as sizeof(a) in main.

19. Consider the following code what will be the output

```
#include <stdio.h>
int main()
{
 int a[5] = {1,2,3,4,5};

 printf("%lu %lu", sizeof(a), sizeof(&a));
}
```

20. How many memory segments are available and explain them.